

A Report on

Smart Parking Assistant System

for

**Self Learning Micro Project of Second Year, Semester III, Course :
Instrumentation and control systems**

in

Electronics & Telecommunication Engineering

by

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Under the guidance of

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**SHAH AND ANCHOR KUTCHHI ENGINEERING
COLLEGE**

Academic Year: 2025-2026



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CERTIFICATE

This is to certify that the project entitled **Micro project Title** is a bonafide work of

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submitted to the Course In charge , Instrumentation and control systems , Shah and Anchor Kutchhi Engineering college , Mumbai in partial fulfillment of the requirement for the award of Self learning **Micro project , Second Year, (SY Sem-III)** in **Electronics & Telecommunication Engineering** during academic year **2025-26**

Dr. Kalpana Chaudhary



1 Title of Micro project : Smart Parking Assistant System

2. ABSTRACT

The Smart Parking Assistant System is an automated solution designed to help drivers quickly locate vacant parking spaces, improving traffic management and reducing congestion. Utilizing Ultrasonic sensors, the system detects the presence of vehicles in parking slots and indicates slot status via LEDs green for vacant and red for occupied. Operating on a 5V DC supply, the system employs a transistor-based circuit to process sensor signals and control LEDs, eliminating the need for microcontrollers.

The methodology involves continuous monitoring of parking slots, real-time processing of detection signals, and immediate visual feedback through LEDs. Expected outcomes include enhanced efficiency, reduction of time spent searching for parking, improved traffic flow, and a cost-effective implementation for small to medium parking areas.

This project demonstrates the practical use of instrumentation and control systems in everyday applications and provides a foundation for scalable smart parking solutions.

3. INTRODUCTION

3.1 Background of Instrumentation & Control Systems

Instrumentation and control systems are fundamental in modern automation technologies, allowing machines to sense, measure, and respond to various parameters automatically. These systems are widely used in industries, traffic management, home automation, and robotics. Integration of sensors, actuators, and control logic enables real-time monitoring and management of processes, increasing safety, precision, and productivity.

3.2 Importance of Automation and Measurement

Automation improves efficiency and reliability by reducing human intervention. Accurate measurement is essential for proper control and decision-making. Automated parking systems provide instant information on slot availability, ensuring smoother vehicle flow and minimizing human errors. They also contribute to energy efficiency, cost reduction, and enhanced user convenience.

3.3 Objectives of the Micro Project

- * Detect vehicle presence in parking slots.**
- * Implement visual indicators using LEDs for slot availability.**
- * Reduce human monitoring in parking areas.**
- * Develop a cost-effective, low-power, and scalable automated system.**
- * Apply instrumentation and control principles in a practical scenario.**
- * Lay a foundation for future IoT-based smart parking solutions.**

4. PROBLEM DEFINITION

4.1 Problem Statement

Finding available parking in congested areas is often time-consuming and frustrating. Manual supervision is inefficient and error-prone. Traffic congestion increases when drivers search for parking, leading to wasted time, fuel consumption, and environmental pollution. An automated parking assistant system can alleviate these problems by providing real-time slot availability.

4.2 Inputs, Process, and Outputs

Inputs: Vehicle presence detection from Ultrasonic sensors.

Process: Ultrasonic sensors emit waves and measure reflections to detect vehicles. Sensor signals are processed through a transistor circuit to control LEDs. Status updates are continuous in real-time.

Outputs: LED indicators display Green for vacant slots and Red for occupied slots, providing immediate visual feedback.

5. LITERATURE REVIEW

Automated parking systems have been developed to optimize space and reduce traffic. Modern solutions use microcontrollers, IoT modules, and cloud monitoring for remote slot tracking. However, these can be costly and require programming expertise. The Smart Parking Assistant System offers a low-cost, simple alternative suitable for small to medium parking areas.

Industry applications include malls, hospitals, offices, airports, and public parking zones. Automation enhances efficiency, reduces the need for attendants, and improves user convenience. Studies show that using Ultrasonic sensors with visual indicators can significantly improve parking management even without microcontrollers.

6. SYSTEM DESIGN

6.1 Block Diagram

The system includes a power supply, Ultrasonic sensor, signal processing unit, and LED indicators. The sensor detects vehicle presence, signals are processed via a transistor driver, and LEDs display the slot status.

6.2 Circuit Description

The Ultrasonic sensor output connects to the transistor base via a resistor. The transistor switches current through the LEDs. Vehicle detection triggers the Red LED, while vacant slots trigger the Green LED, ensuring reliable operation with minimal components.

6.3 Algorithm / Flowchart of Operation

- 1. Power ON system.**
- 2. Ultrasonic sensor emits waves.**
- 3. Vehicle detected → sensor output LOW → Red LED ON, Green LED OFF.**
- 4. No vehicle → sensor output HIGH → Green LED ON, Red LED OFF.**
- 5. Repeat for all parking slots continuously.**

6.4 Major Component Descriptions

Ultrasonic Sensor Module: Detects vehicle presence via wave reflection.

LED Indicators: Display slot status (Green = vacant, Red = occupied).

Transistor BC547: Switches LEDs based on sensor output.

Resistors: Limit current and provide transistor biasing.

5V DC Power Supply: Provides stable voltage for sensors and LEDs.

7. IMPLEMENTATION

7.1 Hardware Used

Ultrasonic Sensor Module

Red and Green LEDs

Transistor BC547

Resistors (220Ω , $10k\Omega$)

5V DC Adapter / Regulated Power Supply

Breadboard or PCB board

Connecting wires

7.2 Software/Tools Used

Circuit simulation software (Proteus / Tinkercad)

Arduino IDE (optional for testing)

7.3 Step-by-Step Working

- 1. Connect Ultrasonic sensor to 5V and GND.**
- 2. Connect sensor output to transistor base via $10k\Omega$ resistor.**
- 3. Connect LEDs to transistor collector via 220Ω resistors.**
- 4. Vehicle enters slot → sensor output LOW → transistor conducts → Red LED ON.**
- 5. Slot vacant → sensor output HIGH → transistor OFF → Green LED ON.**
- 6. System monitors slots continuously.**
- 7. Multiple slots connected in parallel using same principle.**

8. RESULTS & OBSERVATIONS

Observations:

Ultrasonic sensor accurately detected vehicle presence.

Red LED ON when occupied, Green LED ON when vacant.

Consistent performance across multiple tests.

Sensor output: HIGH (~5V) for vacant, LOW (~0V) for occupied.

Over 95% accuracy and stable operation.

Low power consumption.

Research Findings:

Ultrasonic sensors are cost-effective for parking detection.

Transistor circuits can drive LEDs without microcontrollers.

Real-time LED feedback improves parking efficiency.

System reduces human supervision and traffic congestion.

Scalable for small to medium parking areas.

Outdoor conditions may require sensor calibration.

9. ADVANTAGES AND LIMITATIONS

Advantages:

- Low-cost, easy to assemble.**
- Real-time visual indication of slot availability.**
- Low power consumption and maintenance-free.**
- No microcontroller required.**
- Scalable for small to medium parking areas.**

Limitations:

- Affected by extreme sunlight or reflective surfaces.**
- Ultrasonic detection range limited; calibration may be needed.**
- Cannot detect vehicle type or exact dimensions.**
- Large-scale implementation may complicate wiring.**

10. APPLICATIONS

- Malls, shopping centers, office parking lots.**
- Residential complexes and apartments.**
- Airports and railway stations.**
- Hospitals and educational institutions.**
- Smart city and automated parking infrastructure.**
- Traffic monitoring and vehicle counting systems.**

11. CONCLUSION

The Smart Parking Assistant System detects and displays parking slot availability using Ultrasonic sensors and LEDs. It provides a low-cost, reliable, real-time solution for parking management, demonstrating practical applications of automation and control systems to reduce congestion, improve efficiency, and enhance user convenience.

Future Scope:

IoT integration for remote monitoring.

Multi-floor parking management.

Advanced sensors for vehicle type detection.

Data logging and traffic pattern analysis.

12. REFERENCES

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